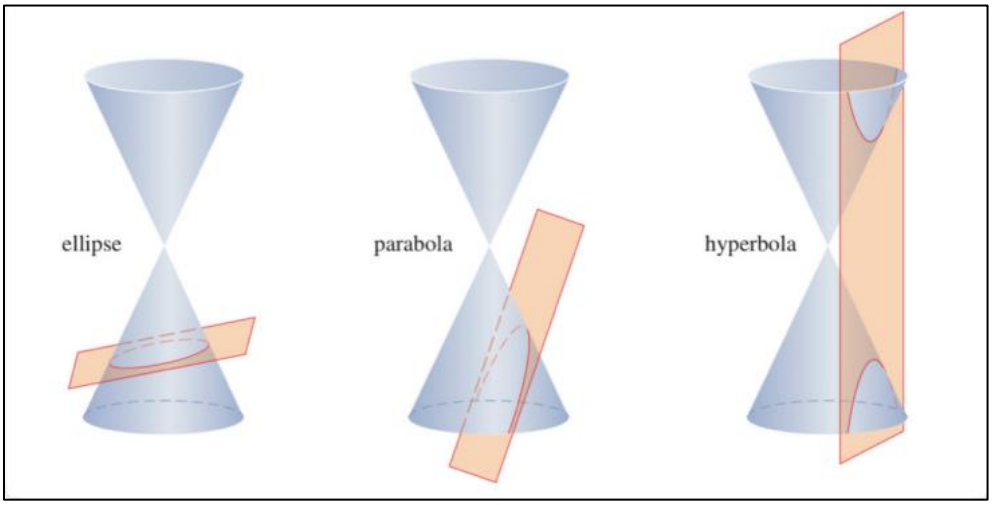
Lecture 19 & 20

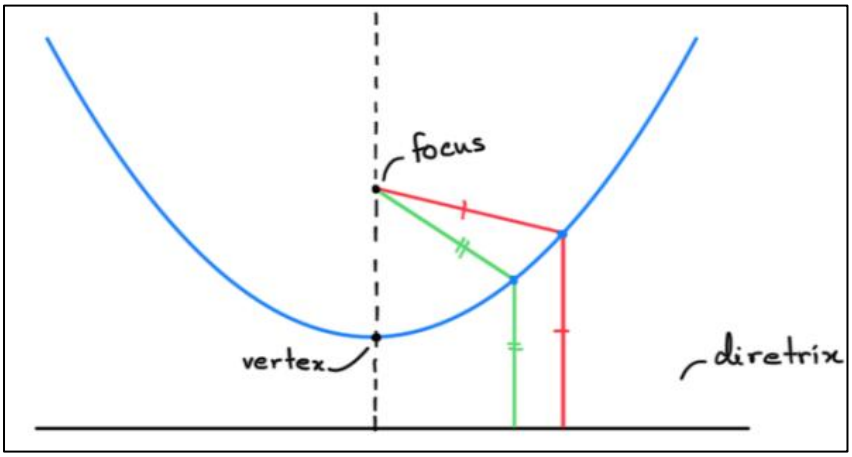
Conics

Parabola, ellipse and hyperbola are called conic section because they result from intersecting a cone with a plane.



Parabola: A parabola is the set of points in a plane that are equidistant from a fixed point and a fixed line.

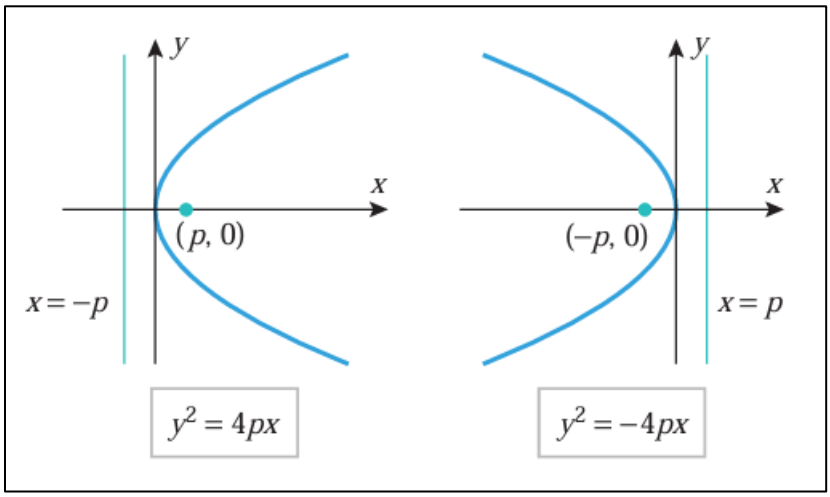
- The point is called the focus of the parabola.
- The line is called the directrix of the parabola.
- The vertex is a point on the parabola which intersects the line which is perpendicular to the directrix and passes through the focus.



Curves shaped like this are called parabola.

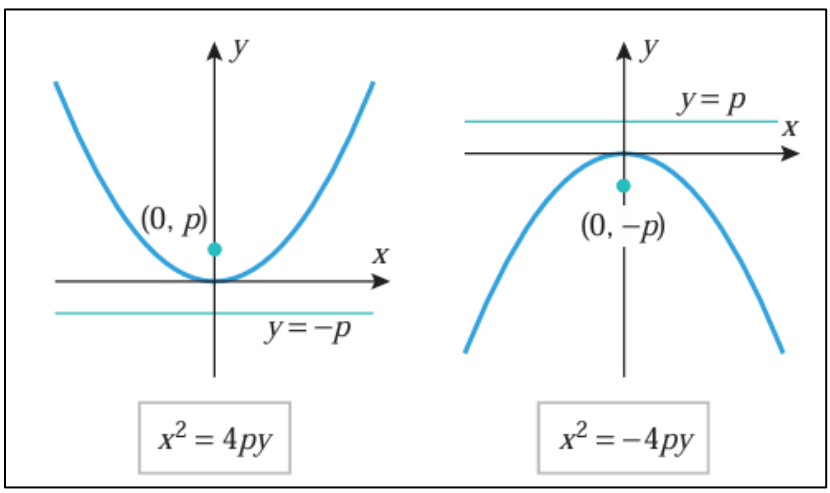
- The dotted vertical line is also called the axis or the axis of symmetry of parabola.

<u>Standard forms of Parabola:</u> (Center/Vertex at (0,0))



⇒ Focal axis, x axis or $y = 0$

(Axis with both focus and vertex/vertices.)



⇒ Focal axis, y axis or $x = 0$

(Axis with both focus and vertex/vertices.)

Here, $(p, 0)$, $(-p, 0)$, $(0, p)$, $(0, -p)$ are focus and $x = -p$, $x = p$, $y = -p$, $y = p$ are the directrix.

- Eccentricity, $e = 1$
- Length of Latus rectum $= |4p|$
- Focal length (distance from center to focus) $= p$

- Parabolas with vertex (h, k) and axis parallel to x-axis,

$$(y-k)^2 = 4p(x-h) \qquad \text{[opens right]}$$

$$(y-k)^2 = -4p(x-h) \qquad \text{[opens left]}$$

- Parabolas with vertex (h, k) and axis parallel to y-axis,

$$(x-h)^2 = 4p(y-k) \qquad \text{[opens up]}$$

$$(x-h)^2 = -4p(y-k) \qquad \text{[opens down]}$$

Example: Find the focus and directrix of the parabola $y^2 + 10x = 0$ and sketch the graph.

Solution: Given,

$y^2 + 10x = 0$

$\Rightarrow\ y^2 = -10x$

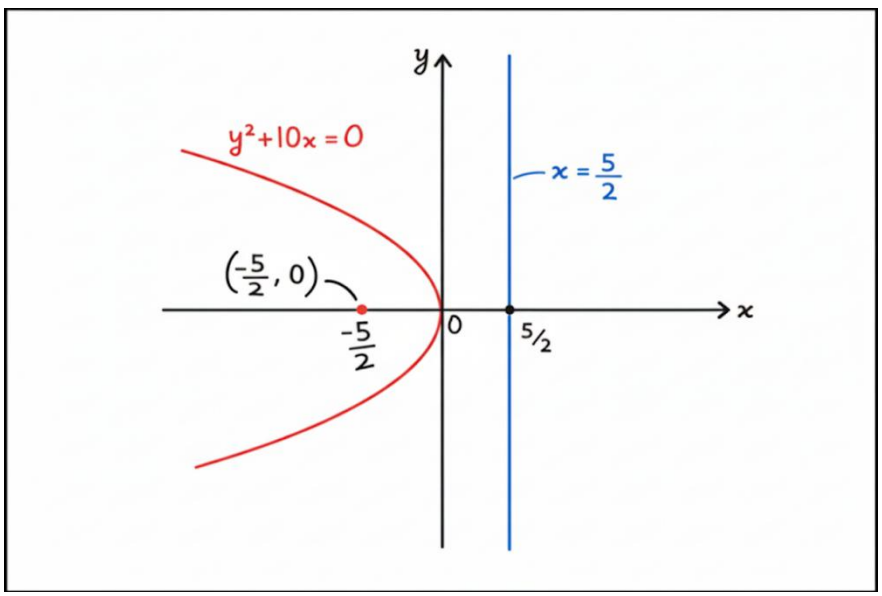
$\Rightarrow\ y^2 = -4\left(\frac{5}{2}\right)x$

Comparing the equation with the standard form, we get, $p = \frac{5}{2}$.

$\therefore$ the focus is $\left(-\frac{5}{2}, 0\right)$

and the directrix is $x = p = \frac{5}{2}$.

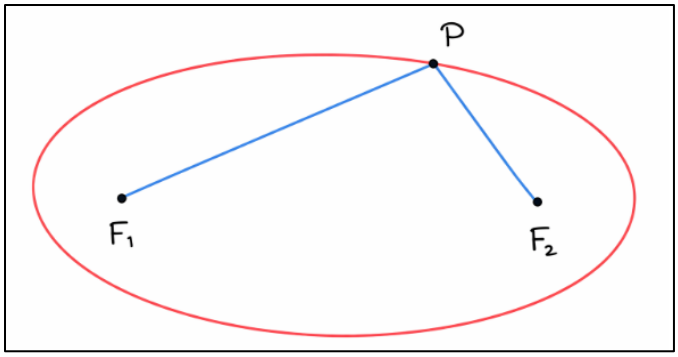
Here, the vertex is (0,0).



Always label the following:

- curve
- focus
- directrix

<u>Ellipse:</u> An ellipse is the set of the points in a plane the sum of whose distance from two fixed points, F_1 and F_2 is a constant. These two fixed points are called the foci (plural of focus).

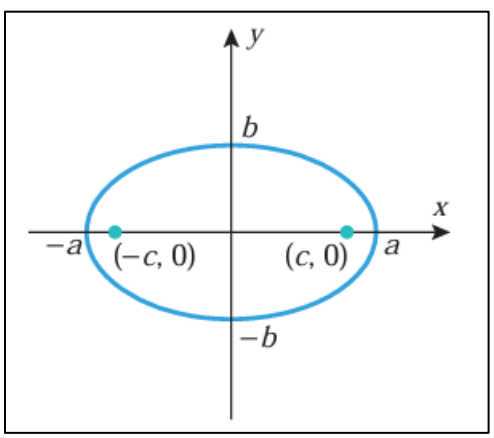


So for any point on ellipse, P:

$$|F_1P| + |F_2P| = \text{constant.}$$

- Major axis: The straight line which passes through both of the foci.
- Center of the ellipse: The point on the major axis that's on the midpoint between the foci.
- Minor axis: The straight line perpendicular to the major axis and passing through the center.
- Vertices: End points of ellipse on major axis.
- Foci always lie on the major axis.

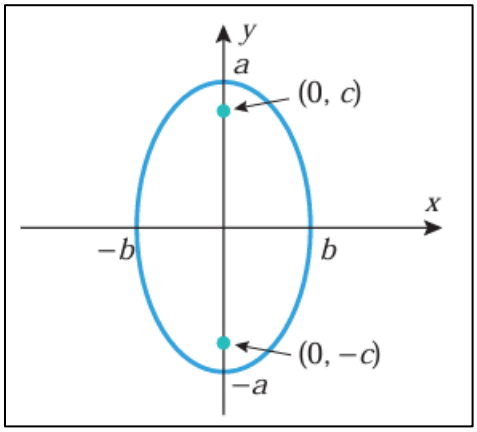
<u>Standard form of Ellipse:</u> (Center at (0,0))



$a \geq b > 0$

- $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$
  - foci $= (\pm c, 0)$
  - vertices $= (\pm a, 0)$

- Equation of directrix, $x = \pm\frac{a^2}{c}$



- $\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$
  - foci $= (0, \pm c)$
  - vertices $= (0, \pm a)$
  - Equation of directrix, $y = \pm\frac{a^2}{c}$

For both of the forms:

- $c^2 = a^2 - b^2$
- Eccentricity, $e = \frac{\sqrt{a^2-b^2}}{a}$
- Length of latus rectum $= \frac{2b^2}{a}$
- Length of major axis $= 2a$
- Length of minor axis $= 2b$

- **Ellipse with center (h, k):**

Horizontal ellipse (major axis parallel to x-axis):

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$$

Vertical ellipse (major axis parallel to y-axis):

$$\frac{(x-h)^2}{b^2} + \frac{(y-k)^2}{a^2} = 1$$

Example: Sketch the graph of $9x^2 + 16y^2 = 144$ and locate the foci.

Solution: Given,

$9x^2 + 16y^2 = 144$

$\Rightarrow \dfrac{9x^2}{144} + \dfrac{16y^2}{144} = 1$

$\Rightarrow \dfrac{x^2}{16} + \dfrac{y^2}{9} = 1$

$\Rightarrow \dfrac{x^2}{4^2} + \dfrac{y^2}{3^2} = 1$

Here, $a = 4$ and $b = 3$.

We know that,

$c^2 = a^2 - b^2$

$\Rightarrow c^2 = 4^2 - 3^2 = 16 - 9$

$\Rightarrow c^2 = 16 - 9$

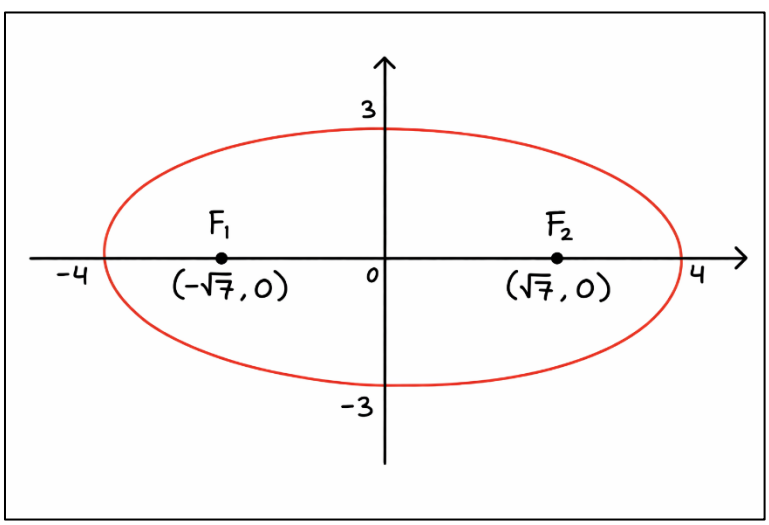
$\Rightarrow c^2 = 7$

$\therefore c = \sqrt{7}$

Here, the major axis is horizontal.

As, the foci lie on the horizontal axis, so, the foci are, $(-\sqrt{7}, 0)$ and $(\sqrt{7}, 0)$.

And the vertices are, $(4,0)$ and $(-4,0)$.



Example: Find the equation of an ellipse with foci $(0, \pm 2)$ and vertices $(0, \pm 3)$.

Solution: Here, the foci are $(0,2)$ and $(0, -2)$.

So, we have, $c = 2$.

Also, the foci are on the vertical axis, so we can say that the major axis is the vertical axis.

The vertices are $(0,3)$ and $(0, -3)$.

So, $a = 3$.

We know,

$c^2 = a^2 - b^2$

$\Rightarrow\ b^2 = a^2 - c^2$

$\Rightarrow\ b^2 = 3^2 - 2^2$

$\Rightarrow\ b^2 = 9 - 4$

$\Rightarrow\ b^2 = 5$

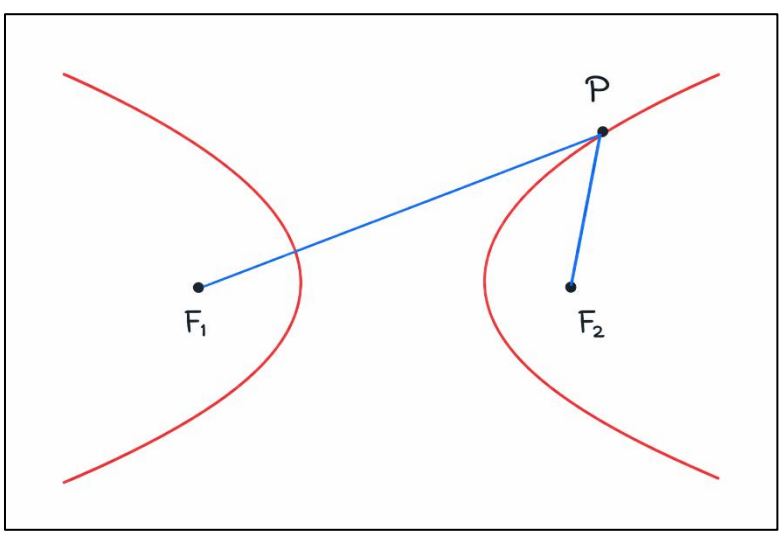
$\therefore\ b = \sqrt{5}$

So, the equation of the ellipse is,

$$\frac{x^2}{\left(\sqrt{5}\right)^2} + \frac{y^2}{3^2} = 1$$

$$\therefore \frac{x^2}{5} + \frac{y^2}{9} = 1$$

<u>Hyperbola:</u> A hyperbola is the set of all points in a plane the difference of whose distance from two fixed points F_1 and F_2 (the foci) is a constant.

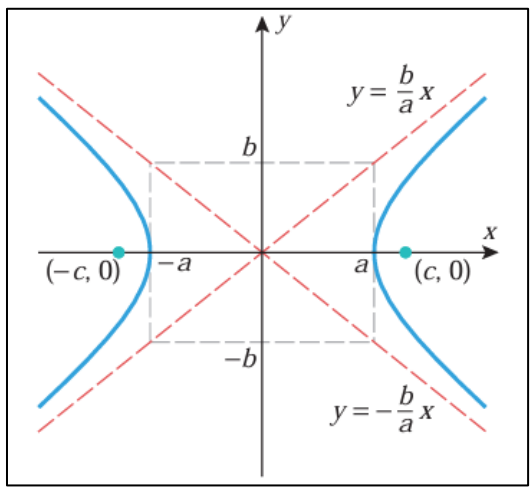


So for any points on hyperbola, P:

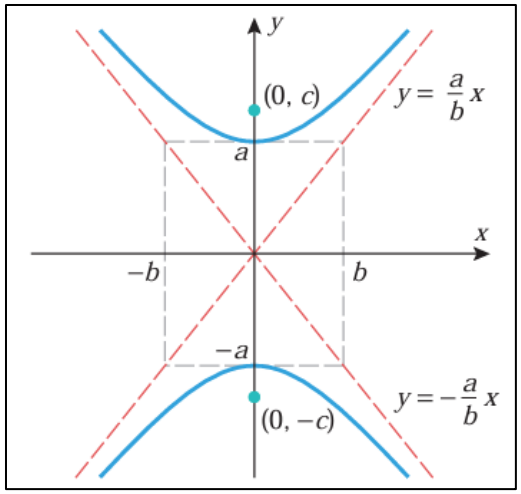
$$|F_1P| - |F_2P| = \text{constant}$$

- Focal Axis: Line segment joining the foci of the hyperbola.
- Center of the Hyperbola: Point on the focal axis of the hyperbola that's at the midpoint between the foci.
- Conjugate Axis: Line perpendicular to the focal axis and passing through the center of the hyperbola.
- Vertices: Intersections of the hyperbola with the focal axes.

<u>Standard form of Hyperbola:</u> (Center at (0,0))



- $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$
  - foci $= (\pm c, 0)$
  - vertices $= (\pm a, 0)$
  - Equation of directrix, $x = \pm\frac{a^2}{c}$
  - Asymptotes, $y = \pm\frac{b}{a}x$



- $\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$
  - foci $= (0, \pm c)$
  - vertices $= (0, \pm a)$
  - Equation of directrix, $y = \pm \frac{a^2}{c}$
  - Asymptotes, $x = \pm \frac{b}{a} y$

For both of the forms:

- $c^2 = a^2 + b^2$
- Length of Latus rectum $= \frac{2b^2}{a}$
- Eccentricity, $e = \frac{\sqrt{a^2+b^2}}{a}$

Hyperbola with center $(\boldsymbol{h}, \boldsymbol{k})$:

Horizontal hyperbola (focal axis parallel to x-axis):

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$$

Vertical hyperbola (focal axis parallel to y-axis):

$$\frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1$$

Example: Find the foci and asymptote of the hyperbola $9x^2 - 16y^2 = 144$. Then sketch the graph.

Solution: Given,

$9x^2 - 16y^2 = 144$

$\Rightarrow \frac{9x^2}{144} - \frac{16y^2}{144} = 1$

$\Rightarrow \frac{x^2}{16} - \frac{y^2}{9} = 1$

$\Rightarrow \frac{x^2}{4^2} - \frac{y^2}{3^2} = 1$

Here, $a = 4$ and $b = 3$.

We know that,

$c^2 = a^2 + b^2$

$\Rightarrow c^2 = 4^2 + 3^2$

$\Rightarrow c^2 = 16 + 9$

$\Rightarrow c^2 = 25$

$\Rightarrow c = \sqrt{25}$

$\therefore c = 5$

Here, the focal axis is horizontal, so the foci lie on the horizontal axis.

$\therefore$ The foci are at $(5,0)$ and $(-5,0)$.

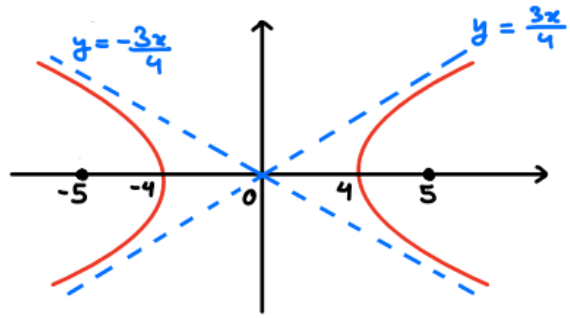
The vertices are also on the horizontal axis. So, the vertices are $(4,0)$ and $(-4,0)$.

The asymptotes are at,

$y = \pm\frac{b}{a}x$

$\Rightarrow y = \pm\frac{3}{4}x$

$\therefore y = \frac{3}{4}x$ and $y = -\frac{3}{4}x$.


$y = -\frac{3x}{4}$
$y = \frac{3x}{4}$
-5
-4
0
4
5